Create and Deploy an Azure Function App Using Visual Studio (Step by Step)

The following steps assume:

- Visual Studio 2022 (or later)
- .NET 7 / .NET 8
- Azure subscription
- Background in ASP.NET Core (C#)

Part 1: Prerequisites

1. Install Required Components

Open **Visual Studio Installer** and ensure:

- **ASP.NET and web development** workload is installed
- **Azure development** workload is installed
- .NET 7 / .NET 8 SDK is installed

Restart Visual Studio after installation.

Part 2: Create Azure Function App in Visual Studio

Step 1: Create a New Project

1. Open **Visual Studio**
2. Click **Create a new project**
3. Search for **Azure Functions**
4. Select **Azure Functions**
5. Click **Next**

Step 2: Configure Project

1. Project name: SampleFunctionApp
2. Location: choose your folder

3. Solution name: same or different
4. Click **Next**

Step 3: Select Runtime & Options

1. **Target Framework**:
    - .NET 8 (Isolated) (Recommended)
    - or .NET 6 / 7 if required
2. **Azure Functions version**: v4
3. **Authorization level**: Function (for HTTP trigger)
4. **Storage account**:
    - Select **Storage Emulator / Azurite** (for local dev)
5. Click **Create**

Visual Studio will generate a Function App project.

Part 3: Understand the Generated Project Structure

Key files:

- Program.cs – entry point (similar to ASP.NET Core)
- local.settings.json – local configuration (not deployed)
- Function1.cs – sample function
- host.json – global function settings

Part 4: Create an HTTP Trigger Function

Step 4: Add New Function

1. Right-click the project → **Add** → **New Azure Function**
2. Select **HTTP trigger**
3. Function name: HelloFunction

4. Authorization: Function
5. Click **Add**

Sample Function Code (C# – Isolated)

```
using Microsoft.Azure.Functions.Worker;
using Microsoft.Azure.Functions.Worker.Http;
using Microsoft.Extensions.Logging;
using System.Net;

public class HelloFunction
{
    private readonly ILogger _logger;

    public HelloFunction(ILoggerFactory loggerFactory)
    {
        _logger = loggerFactory.CreateLogger<HelloFunction>();
    }

    [Function("HelloFunction")]
    public HttpResponseData Run(
        [HttpTrigger(AuthorizationLevel.Function, "get", "post")] HttpRequestData req)
    {
        _logger.LogInformation("HelloFunction triggered");

        var response = req.CreateResponse(HttpStatusCode.OK);
        response.WriteString("Hello from Azure Functions!");
        return response;
```

```
    }
}
```

Part 5: Run Azure Function Locally

Step 5: Start Local Execution

1. Press **F5**
2. Azure Functions Core Tools will start
3. Console window shows the local URL:
4. http://localhost:7071/api/HelloFunction
5. Open browser or Postman and test the endpoint

Part 6: Create Azure Function App in Azure (from Visual Studio)

Step 6: Publish from Visual Studio

1. Right-click the project → **Publish**
2. Select **Azure**
3. Select **Azure Function App (Windows)** or **Linux**
4. Click **Next**

Step 7: Create New Azure Resources

1. Click **Create new**
2. Fill in details:
    - **Function App name**: unique name
    - **Subscription**
    - **Resource Group**: new or existing
    - **Hosting Plan**: Consumption (recommended)
    - **Storage Account**: new or existing

- **Region**

3. Click **Create**

Visual Studio provisions resources and publishes the app.

Part 7: Verify Deployment in Azure Portal

Step 8: Test Function in Azure

1. Go to **Azure Portal**
2. Open **Function App**
3. Click **Functions → HelloFunction**
4. Select **Get Function URL**
5. Test in browser or Postman

Summary Flow

Visual Studio

→ Create Azure Function Project

→ Run locally

→ Publish to Azure

→ Verify in Portal

→ Monitor via App Insights